NON-MF GROUPS

SAUERS

Some groups are not MF. We construct groups for which every homomorphism to an MF group is trivial; the obstruction is a property-(T) subgroup conjugated into a proper subgroup of itself. If a countable unital ring R contains s, t with $ts = 1$ and $R(1 - st)R = R$, then every homomorphism from $\mathrm{EL}_n(R)$ to an MF group is trivial for all $n \geq 2$. In particular, the unit group H of the binary Leavitt algebra $L_{\mathbb{F}_2}(1, 2)$ is a finitely generated simple property-(T) group that is not MF, and $C^*_{\mathrm{r}}(H)$ is separable, stably finite, and not MF. We also construct a sofic non-MF group $W = K \rtimes \mathbb{Z}$ with K locally residually finite, whose canonical trace on $C^*_{\max}(W)$ is amenable but not quasidiagonal, and a two-generated, finitely presented, torsion-free, acylindrically hyperbolic property-(T) group with no nontrivial homomorphism to an MF group.

Draft. This manuscript is under revision.

1. Introduction

In early August 2026 the author was using AI models on two questions raised by the nonsofic group of [OAI]: whether hyperlinearity implies soficity, and whether every group is hyperlinear. In one ChatGPT conversation, GPT-5.6 Sol, on "high" thinking mode, worked on weak MF (matricial field) approximations of a Clifford-twisted wreath product as a step toward hyperlinearity. Earlier in that conversation the author had asked whether to pivot to the MF question, and Sol had advised against it. The author gave Sol the existing progress on the hyperlinear problem and sent this prompt:

> "Achieve f{MF of the Clifford Kun–Thom extension }$\widetilde{W}$ in such a way that it archives our goals and ends everything, or, possibly, a different breakthrough, leading to achieving the final result. You got this!"

Sol answered with an MF counterexample: every norm-matrix model of the Clifford-twisted wreath product kills the Clifford sign. That was on August 9. Also on August 9, the models proved that the maximal group C^*-algebra of a strictly compressed Kazhdan group is infinite. Parallel chats, including one in "Pro" mode, and the agents working on the hyperlinear question, did not find the counterexample. By August 12 the models had reduced the obstruction to one commutator in a finitely presented group and had proved in Lean that it is not MF. They wrote the first draft of this paper that evening.

The author prompted the models, set the direction, retrieved the literature, and suggested the MF question. With the models, the author explored several routes to a non-MF group, shaped the current proof, and checked and revised the citations. The

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models drafted much of the text; the author wrote the rest and edited the whole for clarity and for the mathematical presentation. GPT-5.6 Sol in Codex, and Claude Fable 5 and Claude Opus 5 in Claude Code, wrote the Lean proofs. The whole exchange is in the repository's public commit history.[1]

The author consulted experts from August 13, and their comments shaped the revisions. Many models also reviewed drafts; Astra's reviews shortened the paper and simplified the proof of Theorem 2.7. We recommend Eckhardt's writeup [Ec]: a short, elementary proof that certain generalized wreath products are not MF.

Sol's original counterexample is the group

$$W = \mathrm{Cl}(X) \rtimes V,$$

where V is the ascending HNN extension of $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ along $(v, A) \mapsto (2v, A)$, the set X is the coset space of $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ in V, and $\mathrm{Cl}(X)$ is generated by involutions c_x, $x \in X$, and a central involution ε, with $[c_x, c_y] = \varepsilon$ for $x \neq y$. Every homomorphism from W to an MF group kills ε (Section 4).[2] This paper focuses on a different counterexample,

$$H = L_{\mathbb{F}_2}(1,2)^{\times},$$

the unit group of the binary Leavitt algebra, because H is simple and every homomorphism from H to an MF group is trivial. The right module R_R over $R = L_{\mathbb{F}_2}(1,2)$ satisfies $R_R \cong R_R \oplus R_R$, and H is its automorphism group; Khanh–Thanh show that $H \cong \mathrm{GL}_n(R) = \mathrm{EL}_n(R)$ for every $n \geq 2$ [K–T, Proposition 4.2 and Corollary 4.4]. OpenAI used this group to construct the first nonsofic group [OAI, Chapter 3].

We work with countable groups throughout. A countable group G is *MF* [CDE] if there are positive integers d_n and maps $V_n \colon G \to \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \qquad (g \in G \setminus \{1\}).$$

In other words, G is MF if there are finite unitary matrices, one for each element of G, that multiply correctly up to an error tending to zero in operator norm while the matrices assigned to a nonidentity element do not converge to the identity. For countable G, the second condition may equivalently be required along the full sequence with a constant independent of g [Kor, Propositions 2 and 7]. The strong convergence convention of [GKM, Sch] also requires the models to reproduce the operator norms of the left regular representation, so a group that is not MF as defined here is not MF in that convention either. Equivalently, G embeds in the unitary group of the norm matrix corona

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here and below, products of C^*-algebras mean bounded products, and $\bigoplus$ denotes the ideal of norm-null sequences. The quotient $\mathcal{Q}_{\mathbf{d}}$ is the corona algebra of $\bigoplus_n M_{d_n}(\mathbb{C})$,

[1] https://github.com/SauersML/group-approximation

[2] The Lean proof was registered at Palomar on August 24, 2026: https://palomar-registry.org/entry?id=PALOMAR-2026-08-24-000006&version=1.

since $\prod_n M_{d_n}(\mathbb{C})$ is its multiplier algebra. We call a homomorphism $G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ a *corona homomorphism*. A separable C^*-algebra is MF if it embeds in a norm matrix corona [BK].

We use the group commutator convention $[g,h] = ghg^{-1}h^{-1}$. The groups here fail to be MF because they contain a property-(T) subgroup that is conjugated into a proper subgroup of itself, together with an element that commutes with the subgroup while its conjugate does not. For $L \le G$ put

$$\mathrm{Comp}_G(L) = \{u \in G : uLu^{-1} \le L\}.$$

The elements of $\mathrm{Comp}_G(L)$ are the compressors of L. An element c of the centralizer $C_G(L)$ commutes with L, so ucu^{-1} commutes with uLu^{-1} but need not commute with the rest of L. The normal subgroup of G generated by the commutators of all such elements ucu^{-1} with elements of L is

$$\mathfrak{D}_G(L) = \langle\!\langle [ucu^{-1}, \ell] : u \in \mathrm{Comp}_G(L),\ c \in C_G(L),\ \ell \in L \rangle\!\rangle_G. \tag{1}$$

For every homomorphism $f\colon G \to H$,

$$f(\mathfrak{D}_G(L)) \le \mathfrak{D}_H(f(L)), \tag{2}$$

since compressors, centralizing elements, and commutators of L map to compressors, centralizing elements, and commutators of $f(L)$, and conjugates map to conjugates.

Theorem A (one-sided compression criterion)**.** *Let G be countable and let $L \le G$ have property* (T)*. If $K \trianglelefteq G$ has property* (T) *and $K \le \mathfrak{D}_G(L)$, then every homomorphism from G to an MF group is trivial on K. In particular, if such a K is nontrivial, then G is not MF, and if G has property* (T) *and $\mathfrak{D}_G(L) = G$, then every homomorphism from G to an MF group is trivial.*

Asymptotic multiplicativity in operator norm makes the conjugation maps $\mathrm{Ad}(V_n(g))\colon x \mapsto V_n(g)xV_n(g)^*$ an asymptotically multiplicative family of unitaries on $M_{d_n}(\mathbb{C})$ with its normalized Hilbert–Schmidt inner product. Here $\|\mathrm{Ad}(A) - \mathrm{Ad}(B)\| \le 2\,\|A - B\|$, with the norm on the left taken in the operator algebra of this Hilbert space. The classes of these maps form a homomorphism of G into the unitary group of the norm matrix corona with coordinate sizes d_n^2. Let U be the class of $(\mathrm{Ad}(V_n(u)))$ and P the image of the Kazhdan projection of L in that corona. Then $U^*PU \le P$, because $uLu^{-1} \le L$ has fewer constraints and more fixed vectors, and stable finiteness of that corona forces $U^*PU = P$. So conjugation by u preserves the Hilbert–Schmidt asymptotic commutant of L, and the matrices assigned to an element of $\mathfrak{D}_G(L)$ tend to 1 in Hilbert–Schmidt norm in every operator norm asymptotic representation (Corollary 2.5). Property (T) of K turns this into triviality in operator norm. A corona homomorphism nontrivial on K compresses to a corner where the Kazhdan projection of K vanishes. On that corner, a group-algebra element within $\frac{1}{4}$ of the Kazhdan projection has operator norm below $\frac{1}{4}$, while Hilbert–Schmidt triviality sends its normalized traces to the sum of its coefficients, which exceeds $\frac{3}{4}$ (Theorem 2.7).[3]

[3] A Hilbert–Schmidt bound on the multiplicative defect does not give an operator norm bound on the conjugation maps, so this argument does not apply to sofic or hyperlinear approximations.

The stable letter of an ascending HNN extension of a property-(T) group along a proper self-embedding is a compressor. For a unital ring R, write $e_{ij}(a) = 1 + aE_{ij}$ and let $\mathrm{EL}_n(R)$ be the subgroup of $\mathrm{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$. A pair $s, t \in R$ with $ts = 1$ gives a compressor in the same way: an explicit element $u \in \mathrm{EL}_4(R)$ conjugates $\mathrm{EL}_3(R)$ into itself by $e_{ij}(a) \mapsto e_{ij}(sat)$, and if $1 - st$ generates R as a two-sided ideal, then one commutator $[ucu^{-1}, \ell]$ normally generates all of $\mathrm{EL}_4(R)$, and an embedding of EL_4 into EL_2 carries the conclusion to every $n \geq 2$.

Theorem B (full complementary idempotents)**.** *Let R be a countable unital associative ring. Suppose that $s, t \in R$ satisfy*

$$ts = 1, \qquad R(1 - st)R = R.$$

For every $n \geq 2$, every homomorphism from $\mathrm{EL}_n(R)$ to an MF group is trivial.

Here $R(1 - st)R$ is the two-sided ideal generated by $1 - st$, so the hypothesis says that $\sum_j a_j(1 - st)b_j = 1$ for finitely many $a_j, b_j \in R$; we call such an idempotent $1 - st$ *full*. Equivalently, R contains two disjoint copies of itself: elements v_0, v_1, w_0, w_1 with $w_i v_j = \delta_{ij}$ (Lemma 3.1). A unital ring is *directly finite* if $ts = 1$ implies $st = 1$. The hypothesis holds in every countable simple unital ring that is not directly finite and in every Leavitt algebra $L_k(1, m)$ over a countable field k, $m \geq 2$ (Corollary 3.3). The ring R need not be finitely generated. In the following example $\mathrm{EL}_n(R)$ is moreover simple. Let $L_{\mathbb{F}_2}(1, 2)$ denote the binary Leavitt algebra, the unital $\mathbb{F}_2$-algebra generated by s_0, s_1, t_0, t_1 subject to

$$t_i s_j = \delta_{ij}, \qquad s_0 t_0 + s_1 t_1 = 1. \tag{3}$$

Theorem C (the unit group of the binary Leavitt algebra)**.** *Let $R = L_{\mathbb{F}_2}(1, 2)$ and let $H = R^\times$ be its unit group. Then $H \cong \mathrm{EL}_4(R)$, and H is finitely generated, nontrivial, simple, has property* (T)*, and every homomorphism from H to an MF group is trivial. So H is not MF, and $C^*_{\mathrm{r}}(H)$ is separable and stably finite but not MF, while $C^*_{\max}(H)$ is not finite: it contains a proper isometry.*

Theorem D (an amenable nonquasidiagonal trace)**.** *There is a sofic group $W = K \rtimes \mathbb{Z}$, with K locally residually finite, that is not MF. The canonical trace on $C^*_{\max}(K)$ is quasidiagonal, and the canonical trace on $C^*_{\max}(W)$ is amenable and not quasidiagonal.*

Section 4 constructs W from the affine group $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ and Clifford lamps. The group W is sofic because locally residually finite groups are sofic and soficity passes to extensions with amenable quotient [ES, Theorem 1]. The trace answers the question of Brown [Br, discussion preceding Proposition 3.5.1] and of Schafhauser, Tikuisis and White [STW, Problem X(1)] whether every amenable trace is quasidiagonal. Tikuisis, White, and Winter proved that faithful traces on separable nuclear C^*-algebras satisfying the universal coefficient theorem are quasidiagonal [TWW]; here $C^*_{\max}(W)$ is not nuclear, since W is not amenable. The canonical trace of $C^*_{\max}(K)$ is quasidiagonal, so the crossed product by $\mathbb{Z}$ destroys quasidiagonality of the canonical trace. Since K is a direct limit of residually finite groups, it is MF [Kor, Corollary 10 and Proposition 13]. So MF groups are not closed under semidirect products with $\mathbb{Z}$, which answers a question of Korchagin [Kor, remark following Proposition 12].

Theorem E (a torsion-free finitely presented example)**.** *There is a two-generated, finitely presented, torsion-free, acylindrically hyperbolic group Q with property* (T) *such that every homomorphism from Q to an MF group is trivial. In particular, no nontrivial quotient of Q is MF.*

Section 5 builds Q from Fournier-Facio's torsion-free property-(T) group [FF, §2] and Hull's small cancellation theorem [Hu].

Related work. Blackadar and Kirchberg introduced MF algebras and proved that a separable C^*-algebra is NF if and only if it is nuclear and MF [BK]. Whether every separable stably finite C^*-algebra is MF, the MF problem [GH, §6], was answered negatively through the failure of the Connes embedding problem [JNVWY][GH, Proposition 6.1 and Remark 6.2]. Theorem C gives a counterexample among reduced group C^*-algebras. Whether every countable group is MF was a longstanding open problem [Kor, BDL].

Theorem A uses the configuration of the nonsofic construction [OAI, Proposition 2.3]: a property-(T) subgroup L conjugated into itself by an element u, and an element c commuting with L whose conjugate ucu^{-1} does not. There, soficity and the rigidity theorems of Kun [Ku16] and Kun–Thom [KT19] force the centralizing subgroup to be locally embeddable into finite groups, and a copy of Thompson's group V gives the contradiction. Here the rigidity comes from the Kazhdan projection of L in a norm matrix corona, which commutes with the compressor because the corona is stably finite, and the contradiction is a commutator $[ucu^{-1}, \ell] \neq 1$ that every operator norm asymptotic representation sends to 1 in Hilbert–Schmidt norm. The group H is the group of [OAI], and Theorem E reuses Fournier-Facio's configuration [FF]. Bachner–Dogon–Lubotzky showed that for groups of Deligne type, operator–Hilbert–Schmidt stability would imply that the group is not MF [BDL, Proposition 1.5]. Here the compression relation replaces the stability hypothesis.

Leavitt introduced the algebras $L_k(1, n)$ [Lev]. The binary Leavitt algebra is purely infinite simple [AA], is an exchange ring [Ara], and has center $\mathbb{F}_2$ [AC]. Preusser's sandwich theorem then gives simplicity of H [Pre], and property (T) is Ershov–Jaikin-Zapirain's theorem [EJZ, Theorem 1.1].

2. One-sided compression

2.1. Corona homomorphisms.

Lemma 2.1. *Let G be countable and $K \leq G$. Every homomorphism from G to an MF group is trivial on K if and only if every corona homomorphism from G is trivial on K.*

Proof. The image of a corona homomorphism is a countable MF group. Conversely, compose a homomorphism to an MF group with a corona embedding of its image; the resulting corona homomorphism has the same kernel. □

2.2. Kazhdan transport in normalized Hilbert–Schmidt norm.

For an actual representation $\rho\colon G \to \mathcal{U}(d)$ and $u \in G$ with $uLu^{-1} \leq L$, the commutant $\mathcal{C} = \rho(L)' \subseteq M_d(\mathbb{C})$ satisfies $\rho(u)^*\mathcal{C}\rho(u) \subseteq \mathcal{C}$, and the two spaces have the same finite dimension, so the inclusion is an equality; if c commutes with L, then $\rho(ucu^{-1})$ commutes with $\rho(L)$, and ρ is trivial on $\mathfrak{D}_G(L)$. An asymptotic representation has no exact commutant to count. Property (T) supplies a Kazhdan projection in its place, and stable finiteness of the corona replaces the dimension count.

For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \mathrm{tr}_d(a^*a)^{1/2}, \qquad \mathrm{tr}_d = \frac{1}{d}\,\mathrm{Tr}\,.$$

An *operator norm asymptotic representation* of a group G is a sequence of maps $V_n \colon G \to \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G).$$

The elements g with $\|V_n(g) - 1\|_2 \to 0$ form a normal subgroup of G, because $\|a\|_2 \le \|a\|$, the Hilbert–Schmidt norm is invariant under adjoints and unitary conjugation, and $\|V_n(g^{-1}) - V_n(g)^*\| \to 0$:

$$\|V_n(gh) - 1\|_2 \le \|V_n(g) - 1\|_2 + \|V_n(h) - 1\|_2 + o(1),$$
$$\left\|V_n(g^{-1}) - 1\right\|_2 = \|V_n(g) - 1\|_2 + o(1), \qquad \left\|V_n(ghg^{-1}) - 1\right\|_2 = \|V_n(h) - 1\|_2 + o(1).$$

For $L \le G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \;\middle|\; \begin{array}{l} \sup_n \|x_n\|_2 < \infty, \\ \|V_n(\ell)x_n - x_nV_n(\ell)\|_2 \longrightarrow 0 \quad (\ell \in L) \end{array} \right\}$$

be the Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\mathrm{Ad}(V(g))$, of the Hilbert–Schmidt bounded matrix sequences. A unital C^*-algebra is finite if each of its isometries is unitary, and stably finite if all matrix algebras over it are finite.

Lemma 2.2. *For every sequence of positive integers (m_n), the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Write A for the displayed corona. If $v^*v = 1$ in A and (x_n) is a bounded lift of v, then $\|x_n^*x_n - 1\| \to 0$, so for all large n the matrix $u_n = x_n(x_n^*x_n)^{-1/2}$ is unitary and $\|u_n - x_n\| \to 0$; after assigning arbitrary unitary values to the finitely many remaining coordinates, (u_n) is a unitary lift of v, and $vv^* = 1$. So A is finite, and so is $M_k(A)$, which is the corona with matrix sizes km_n. Finally, if $p \le q$ are equivalent projections in a finite algebra B and w is a partial isometry with $w^*w = q$ and $ww^* = p$, then $w = qwq$, so $w + (1-q)$ is an isometry of B and hence unitary; therefore $p + (1-q) = (w + (1-q))(w + (1-q))^* = 1$ and $p = q$. $\square$

For a group L with property (T), the Kazhdan projection is the central projection $e_L \in C^*_{\max}(L)$ whose image in every unitary representation of L is the orthogonal projection onto the L-invariant vectors [AW].

Lemma 2.3 (one-sided order for the Kazhdan projection)**.** *Let L have property* (T)*, let B be a unital C^*-algebra, and let $\pi\colon L \to \mathcal{U}(B)$ be a homomorphism. Let $P \in B$ be the image of e_L under the $*$-homomorphism $C^*_{\max}(L) \to B$ induced by π. If a unitary $U \in B$ satisfies $U\pi(L)U^* \subseteq \pi(L)$, then $U^*PU \le P$.*

Proof. Fix a faithful nondegenerate representation of B on a Hilbert space $\mathcal{H}$, and let $\operatorname{Fix}\pi(L)$ be the space of vectors fixed by $\pi(L)$. Then P is the orthogonal projection onto $\operatorname{Fix}\pi(L)$. If $\xi \in \operatorname{Fix}\pi(L)$ and $\ell \in L$, then $\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi$, because $U\pi(\ell)U^*$ belongs to $\pi(L)$. So $U^* \operatorname{Fix}\pi(L) \subseteq \operatorname{Fix}\pi(L)$. Since U^*PU is the orthogonal projection onto $U^* \operatorname{Fix}\pi(L)$, it follows that $U^*PU \le P$. ☐

Theorem 2.4 (one-sided Kazhdan transport)**.** *Let $L \le G$ have property* (T)*, and let $u \in G$ satisfy $uLu^{-1} \le L$. Let (V_n) be an operator norm asymptotic representation of G. Then*

$$\operatorname{Ad}(V(u))\big(\mathcal{C}_2(V,L)\big) = \mathcal{C}_2(V,L).$$

Proof. For unitaries A, B,

$$\|\operatorname{Ad}(A) - \operatorname{Ad}(B)\| \le 2\,\|A - B\|\,,$$

where the norm on the left is the operator norm on the Hilbert space $(M_{d_n}(\mathbb{C}), \|\cdot\|_2)$, so the maps $\operatorname{Ad}(V_n(g))$ are asymptotically multiplicative in operator norm and

$$\widetilde{\sigma}(g) = [\operatorname{Ad}(V_n(g))]_n \in \mathcal{U}(\mathcal{B}), \qquad \mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C}))/\bigoplus_n B(M_{d_n}(\mathbb{C})),$$

is a homomorphism, where $B(M_{d_n}(\mathbb{C}))$ denotes the operators on the Hilbert–Schmidt Hilbert space. The algebra $\mathcal{B}$ is a norm matrix corona with coordinate sizes d_n^2 after a choice of matrix units. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection of L [AW] under $C^*_{\max}(L) \to \mathcal{B}$, and lift P to orthogonal projections P_n on $(M_{d_n}(\mathbb{C}), \|\cdot\|_2)$ by functional calculus.

A Hilbert–Schmidt bounded sequence (x_n) lies in $\mathcal{C}_2(V,L)$ if and only if

$$\|P_n x_n - x_n\|_2 \longrightarrow 0.$$

In one direction, fix $\varepsilon > 0$ and, by density of the group algebra in $C^*_{\max}(L)$, a finite set $F \subseteq L$ and scalars $(a_\ell)_{\ell\in F}$ with

$$\Big\|\sum_{\ell\in F} a_\ell u_\ell - e_L\Big\| < \varepsilon,$$

where u_ℓ is the canonical unitary of ℓ; the trivial character gives $\big|\sum_{\ell\in F} a_\ell - 1\big| < \varepsilon$. The element $\sum_{\ell\in F} a_\ell \widetilde{\sigma}(\ell)$ of $\mathcal{B}$ is within ε of P, so

$$\limsup_n \left\| P_n - \sum_{\ell\in F} a_\ell \operatorname{Ad}(V_n(\ell)) \right\| \le \varepsilon.$$

If $(x_n) \in \mathcal{C}_2(V,L)$ and $\sup_n \|x_n\|_2 \le c$, then $\operatorname{Ad}(V_n(\ell))x_n = x_n + o(1)$ in $\|\cdot\|_2$ for each $\ell \in F$, so

$$\limsup_n \|P_n x_n - x_n\|_2 \le \varepsilon c + \Big|\sum_{\ell\in F} a_\ell - 1\Big| c \le 2\varepsilon c,$$

and ε was arbitrary. In the other direction, $u_\ell e_L = e_L$ in $C^*_{\max}(L)$, so $\widetilde{\sigma}(\ell)P = P$ and $\|\mathrm{Ad}(V_n(\ell))P_n - P_n\| \to 0$; if $\|P_n x_n - x_n\|_2 \to 0$, then

$$\begin{aligned}\|\mathrm{Ad}(V_n(\ell))x_n - x_n\|_2 &\le 2\,\|x_n - P_n x_n\|_2\\ &\qquad + \|\mathrm{Ad}(V_n(\ell))P_n - P_n\|\ \|x_n\|_2 \longrightarrow 0.\end{aligned}$$

Put $U = \widetilde{\sigma}(u)$. Since $uLu^{-1} \le L$, $U\widetilde{\sigma}(L)U^* \subseteq \widetilde{\sigma}(L)$, so $U^*PU \le P$ by Lemma 2.3, and $U^*PU = P$ by Lemma 2.2, since the two projections are unitarily equivalent. So $[U, P] = 0$ and $\|[\mathrm{Ad}(V_n(u)), P_n]\| \to 0$. For $(x_n) \in \mathcal{C}_2(V, L)$,

$$\begin{aligned}\left\|P_n\,\mathrm{Ad}(V_n(u))^{\pm1}x_n - \mathrm{Ad}(V_n(u))^{\pm1}x_n\right\|_2 &\le \left\|[P_n, \mathrm{Ad}(V_n(u))^{\pm1}]\right\|\ \|x_n\|_2\\ &\qquad + \|P_n x_n - x_n\|_2 \longrightarrow 0,\end{aligned}$$

so $\mathrm{Ad}(V(u))^{\pm1}x \in \mathcal{C}_2(V, L)$ by the displayed equivalence. □

Corollary 2.5. *Let $L \le G$ have property* (T) *and let (V_n) be an operator norm asymptotic representation of G. Then*

$$\|V_n(d) - 1\|_2 \longrightarrow 0 \qquad (d \in \mathfrak{D}_G(L)).$$

Proof. Let $u \in \mathrm{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, and write $\mathcal{C}_2 = \mathcal{C}_2(V, L)$. Since c commutes with L, $(V_n(c)) \in \mathcal{C}_2$, so $(V_n(u)V_n(c)V_n(u)^*) \in \mathcal{C}_2$ by Theorem 2.4, and then $(V_n(ucu^{-1})) \in \mathcal{C}_2$ because $\|V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1})\| \to 0$. So $\|V_n(\ell)V_n(ucu^{-1}) - V_n(ucu^{-1})V_n(\ell)\|_2 \to 0$, and by asymptotic multiplicativity $\|V_n([ucu^{-1}, \ell]) - 1\|_2 \to 0$. The elements with this property form a normal subgroup of G, so it contains $\mathfrak{D}_G(L)$. □

2.3. **From Hilbert–Schmidt to operator norm.** Restricting a corona homomorphism ρ to a corner requires a correction: a projection q commuting with $\rho(G)$ has lifts q_n that commute only asymptotically with unitary lifts $U_n(g)$ of $\rho(g)$, so the compressions $q_n U_n(g) q_n$ are only approximately unitary in the matrix corners.

Lemma 2.6 (central corona corners)**.** *Let $\rho\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$, integers $r_n = \mathrm{rank}(q_n)$, unitary identifications $J_n\colon \mathbb{C}^{r_n} \to q_n\mathbb{C}^{d_n}$, and an operator norm asymptotic representation*

$$W_n\colon G \longrightarrow \mathcal{U}(r_n)$$

with $W_n(1) = I_{r_n}$ such that, in the corona over the retained coordinates, the class of $(J_n W_n(g) J_n^*)$ is the coordinate restriction of $q\rho(g)$, a unitary of the corner $q\mathcal{Q}_{\mathbf{d}}q$ with unit q.

Proof. Lift q to projections $q_n \in M_{d_n}(\mathbb{C})$ by functional calculus. Since $q \neq 0$, infinitely many q_n are nonzero; retain those coordinates and put $r_n = \mathrm{rank}(q_n)$. Unitary identifications $J_n\colon \mathbb{C}^{r_n} \to q_n\mathbb{C}^{d_n}$ identify each corner $q_n M_{d_n}(\mathbb{C}) q_n$ with $M_{r_n}(\mathbb{C})$, and then the corner $q\mathcal{Q}_{\mathbf{d}}q$ with the corona $\mathcal{Q}_{\mathbf{r}}$: a bounded sequence (z_n) representing $z \in \mathcal{Q}_{\mathbf{d}}$ gives the representative $(q_n z_n q_n)$ of qzq. Since q commutes with $\rho(G)$, the map $g \mapsto q\rho(g)$ is a homomorphism from G to the unitary group of the corner, with unit q. Each value is a unitary of $\mathcal{Q}_{\mathbf{r}}$, so, as in the proof of Lemma 2.2, it lifts to unitaries $W_n(g) \in \mathcal{U}(r_n)$, with $W_n(1) = I_{r_n}$. Multiplicativity in the corona makes (W_n) an operator norm asymptotic representation, and the class of $(J_n W_n(g) J_n^*)$ is the coordinate restriction of $q\rho(g)$. □

Theorem 2.7 (normal Kazhdan subgroups)**.** *Let G be countable and let $K \trianglelefteq G$ have property* (T)*. If every operator norm asymptotic representation (V_n) of G satisfies $\|V_n(k) - 1\|_2 \to 0$ for all $k \in K$, then every corona homomorphism from G is trivial on K.*

Proof. Suppose that a corona homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K. Let $e_K \in C^*_{\max}(K)$ be the Kazhdan projection of K, and let $p \in \mathcal{Q}_{\mathbf{d}}$ be its image under the homomorphism $C^*_{\max}(K) \to \mathcal{Q}_{\mathbf{d}}$ induced by $\Theta|_K$. In any faithful representation of the corona, p is the projection onto the K-fixed vectors. Since $gKg^{-1} = K$, the range of $\Theta(g)p\Theta(g)^*$ is $\Theta(g)\operatorname{Fix}\Theta(K) = \operatorname{Fix}\Theta(gKg^{-1}) = \operatorname{Fix}\Theta(K)$, so

$$\Theta(g)p\Theta(g)^* = p \qquad (g \in G),$$

and $q = 1-p$ is nonzero because Θ is nontrivial on K. By Lemma 2.6 there is an operator norm asymptotic representation $W_n\colon G \to \mathcal{U}(r_n)$ whose corona homomorphism $\widehat{\Theta}$ is the coordinate restriction of $g \mapsto q\Theta(g)$. Let $\pi\colon C^*_{\max}(K) \to \mathcal{Q}_{\mathbf{r}}$ be the homomorphism induced by $\widehat{\Theta}|_K$. Since q commutes with $\Theta(K)$, the map $a \mapsto q\Theta(a)$ is a homomorphism on $C^*_{\max}(K)$ agreeing with π on K, so $\pi(e_K)$ is the coordinate restriction of $qp = 0$.

The group algebra $\mathbb{C}[K]$ is dense in $C^*_{\max}(K)$, so there are a finite set $F \subseteq K$ and scalars $(\alpha_k)_{k\in F}$ with

$$a = \sum_{k\in F} \alpha_k u_k, \qquad \|a - e_K\| < \tfrac{1}{4},$$

where u_k is the canonical unitary of k. The trivial character χ of K, the state of $C^*_{\max}(K)$ with $\chi(u_k) = 1$ for every k, has $\chi(e_K) = 1$, so

$$\Big|\sum_{k\in F} \alpha_k - 1\Big| = |\chi(a - e_K)| < \tfrac{1}{4}.$$

Since $\pi(e_K) = 0$,

$$\limsup_n \Big\|\sum_{k\in F} \alpha_k W_n(k)\Big\| = \|\pi(a)\| = \|\pi(a - e_K)\| < \tfrac{1}{4}.$$

The hypothesis applies to (W_n) itself, so

$$|\operatorname{tr}_{r_n}(W_n(k)) - 1| \le \|W_n(k) - I_{r_n}\|_2 \longrightarrow 0 \qquad (k \in K),$$

with the normalized Hilbert–Schmidt norm of $M_{r_n}(\mathbb{C})$, and then

$$\operatorname{tr}_{r_n}\Big(\sum_{k\in F} \alpha_k W_n(k)\Big) \longrightarrow \sum_{k\in F} \alpha_k.$$

Since $|\operatorname{tr}_{r_n}(x)| \le \|x\|$, the limit has absolute value at most $\frac{1}{4}$, while $|\sum_{k\in F} \alpha_k| > \frac{3}{4}$, a contradiction. □

Proof of Theorem A. By Corollary 2.5, every operator norm asymptotic representation of G satisfies $\|V_n(k) - 1\|_2 \to 0$ for $k \in K \leq \mathfrak{D}_G(L)$. By Theorem 2.7, every corona homomorphism is then trivial on K, and by Lemma 2.1 so is every homomorphism to an MF group. If $K \neq 1$, then G does not embed in an MF group, so G is not MF. The last assertion is the case $K = G = \mathfrak{D}_G(L)$. □

2.4. The maximal group C^*-algebra.

Lemma 2.8 (strictly dominated projections). *Let A be a unital C^*-algebra, let $p \in A$ be a projection, and let $u \in A$ be a unitary with $p < upu^*$. Then $s = pu^* + (1 - upu^*)$ is an isometry with $ss^* \neq 1$. In particular A is not stably finite, and no tracial state on A is faithful.*

Proof. Put $q = upu^*$, so that $pq = qp = p$, $up(1-q) = 0$, and $pu^*(1-q) = pu^* - pu^*upu^* = 0$. Then

$$\begin{aligned} s^*s &= upu^* + up(1-q) + (1-q)pu^* + (1-q) = q + (1-q) = 1, \\ ss^* &= pu^*up + pu^*(1-q) + (1-q)up + (1-q) \\ &= p + (1-q) = 1 - (q-p) \neq 1. \end{aligned}$$

A unital C^*-algebra containing a nonunitary isometry is not finite, and neither is any matrix algebra over it, since $\mathrm{diag}(s, 1, \dots, 1)$ is a nonunitary isometry in $M_k(A)$. A tracial state τ has $\tau(q-p) = \tau(s^*s) - \tau(ss^*) = 0$ with $q-p$ a nonzero positive element, so no tracial state on A is faithful. □

Proposition 2.9 (proper one-sided compression). *Let G be a countable group containing a property-*(T) *subgroup Γ and an element t with $t\Gamma t^{-1} \subsetneq \Gamma$. Then $C^*_{\max}(G)$ contains a proper isometry. In particular, $C^*_{\max}(G)$ is not stably finite, admits no faithful tracial state, and is neither residually finite-dimensional nor MF.*

Proof. Let $P \in C^*_{\max}(G)$ be the image of the Kazhdan projection of Γ [AW] under the canonical map $C^*_{\max}(\Gamma) \to C^*_{\max}(G)$, and let u be the canonical unitary of t. Conjugation by u implements the isomorphism $\Gamma \to t\Gamma t^{-1}$ on canonical unitaries, so by Lemma 2.3 applied with $U = u$, $P \leq uPu^*$, where uPu^* is the image of the Kazhdan projection of $t\Gamma t^{-1}$. The inequality is strict: in the quasi-regular representation of G on $\ell^2(G/t\Gamma t^{-1})$, the point mass at the base coset is fixed by $t\Gamma t^{-1}$ and moved by every element of $\Gamma \setminus t\Gamma t^{-1}$, so the two fixed-space projections differ. By Lemma 2.8, $C^*_{\max}(G)$ contains a proper isometry, is not stably finite, and has no faithful tracial state. MF algebras are stably finite, and a separable residually finite-dimensional C^*-algebra is MF [BK]. □

A group can satisfy the hypothesis of Proposition 2.9 and be MF. The ascending HNN extension V of $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ along $(v, A) \mapsto (2v, A)$, used in Section 4, has the faithful matrix realization

$$V \cong \left\{ \begin{pmatrix} 2^k A & v \\ 0 & 1 \end{pmatrix} : A \in \mathrm{SL}_3(\mathbb{Z}),\ v \in \mathbb{Z}[1/2]^3,\ k \in \mathbb{Z} \right\},$$

and reduction modulo odd integers separates its elements, so V is residually finite and MF [Kor, Corollary 10]. So the lamps of Section 4 are necessary for the non-MF conclusion there.

2.5. **A finite certificate.**

Proposition 2.10 (linear collapse)**.** *Let $G = \langle g_1, \dots, g_m\rangle$ be a finitely generated group such that every homomorphism from G to an MF group is trivial. Then there are finitely many words $r_1, \dots, r_s$ in the free group F_m with $r_j(g_1, \dots, g_m) = 1$ in G, and a constant C, such that all unitaries $U_1, \dots, U_m \in \mathcal{U}(d)$, for every $d \geq 1$, satisfy*

$$\max_i \|U_i - 1\| \leq C \max_j \|r_j(U_1, \dots, U_m) - 1\| .$$

Conversely, if such words and such a constant exist and G is countable, then every homomorphism from G to an MF group is trivial.

Proof. The group G is perfect: a nontrivial abelianization of a finitely generated group has a nontrivial finite quotient, and finite groups are MF. So there are words $a_{ik}, b_{ik} \in F_m$ with

$$g_i = \prod_{k=1}^{q_i} [a_{ik}(g_1, \dots, g_m), b_{ik}(g_1, \dots, g_m)] \qquad (1 \leq i \leq m),$$

and the words $w_i = x_i^{-1} \prod_{k=1}^{q_i} [a_{ik}, b_{ik}]$ are relations of G. For a tuple $U = (U_1, \dots, U_m)$ of unitaries put $D(U) = \max_i \|U_i - 1\|$. A word a of length $|a|$ has $\|a(U) - 1\| \leq |a|\, D(U)$, and unitaries A, B satisfy

$$\|[A, B] - 1\| = \|AB - BA\| \leq 2\, \|A - 1\|\, \|B - 1\| ,$$

since $AB - BA = (A-1)(B-1) - (B-1)(A-1)$. So there is a constant B_0, determined by the chosen words, with

$$D(U) \leq \max_i \|w_i(U) - 1\| + B_0\, D(U)^2. \tag{4}$$

Enumerate the relations of G as $r_1, r_2, \dots$ with $w_1, \dots, w_m$ first, and put $\mathcal{R}_n = \{r_1, \dots, r_n\}$. If no $\mathcal{R}_n$ works with the constant n, there are d_n and tuples $U^{(n)}$ in $\mathcal{U}(d_n)$ with

$$D(U^{(n)}) > n\, \delta_n, \qquad \delta_n = \max_{r \in \mathcal{R}_n} \left\| r(U^{(n)}) - 1 \right\| .$$

Since $D(U^{(n)}) \leq 2$, $\delta_n \to 0$, so $g_i \mapsto [U_i^{(n)}]_n$ is a homomorphism from G to the unitary group of $\prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C})$, trivial by hypothesis; so $D(U^{(n)}) \to 0$. For $n \geq m$, the inequality (4) gives $D(U^{(n)}) \leq \delta_n + B_0\, D(U^{(n)})^2$, and division by $D(U^{(n)})$ gives $1 \leq \frac{1}{n} + B_0\, D(U^{(n)})$, which fails for large n.

Conversely, a corona homomorphism has unitary coordinate lifts whose relation defects tend to zero, so the displayed inequality forces every generator lift to the identity, and the homomorphism is trivial; Lemma 2.1 extends this to every homomorphism to an MF group. □

With $r_1, \dots, r_s$ as in Proposition 2.10, put $P = \langle x_1, \dots, x_m \mid r_1, \dots, r_s\rangle$. The inequality holds verbatim for P, so every finitely generated group all of whose homomorphisms to MF groups are trivial is a quotient of a finitely presented group with the same property and the same number of generators. The set of m-marked groups satisfying $r_1 = \cdots = r_s = 1$ is open and closed in the space of marked groups, and every group in it has the property; so the property is open. For the two-generated group Q of

Theorem E, the inequality bounds $\max\{\|U-1\|,\|V-1\|\}$ by the defects of finitely many words in two unitary matrices, uniformly over the matrix size, while the left regular representation solves the same equations exactly with $U,V\neq 1$.

3. One-sided inverses and elementary groups

The proof of Theorem B runs in rank four, with three coordinates for the subgroup EL_3, which has property (T) over a finitely generated ring, and one for its centralizer; two lemmas then carry the conclusion to rank two. The compressor needs only $ts=1$; the ideal condition on $1-st$ enters in the last step of the rank-four argument and in the first lemma.

Lemma 3.1 (two copies)**.** *Let R be a unital ring with $s,t\in R$ such that $ts=1$ and $R(1-st)R=R$. Then there are $v_0,v_1,w_0,w_1\in R$ with $w_iv_j=\delta_{ij}$. Conversely, such elements satisfy $w_0v_0=1$ and $w_1(1-v_0w_0)v_1=1$.*

Proof. Put $e=1-st$, so that $es=te=0$, and choose $a_j,b_j\in R$, $0\le j<m$, with $\sum_j a_jeb_j=1$. Put

$$v_0=s^m,\qquad w_0=t^m,\qquad v_1=\sum_j s^jeb_j,\qquad w_1=\sum_j a_jet^j.$$

For all $i,j\ge 0$, $et^is^je=\delta_{ij}e$: for $j>i$ the middle factor is s^{j-i}, which e kills on the left, and for $i>j$ it is t^{i-j}, which e kills on the right. So $w_0v_0=t^ms^m=1$, $w_1v_1=\sum_j a_jeb_j=1$, $w_0v_1=\sum_j t^{m-j}eb_j=0$, and $w_1v_0=\sum_j a_jes^{m-j}=0$. Conversely, $w_1(1-v_0w_0)v_1=w_1v_1-(w_1v_0)(w_0v_1)=1$. □

Lemma 3.2 (normal generation in rank two)**.** *Let R be a unital ring and let $v,w,a,b\in R$ satisfy $wv=1$, $ba=1$, and $bv=0$. Then*

$$D=[e_{12}(v),e_{21}(b)]=\mathrm{diag}(1+vb,1)$$

normally generates $\mathrm{EL}_2(R)$.

Proof. The matrix $e_{12}(v)e_{21}(b)e_{12}(-v)$ has rows $(1+vb,\,-vbv)$ and $(b,\,1-bv)$, so for $bv=0$ it is $\left(\begin{smallmatrix}1+vb&0\\ b&1\end{smallmatrix}\right)$, and multiplying by $e_{21}(-b)$ gives $D=\mathrm{diag}(1+vb,1)$. Let N be the normal closure of D in $\mathrm{EL}_2(R)$. For $r\in R$, $[D,e_{12}(ar)]=e_{12}((1+vb)ar-ar)=e_{12}(vr)$, since $ba=1$; so $e_{12}(vR)\le N$. Put $f=1-vw$, an idempotent with $fv=0=wf$, and

$$z=vf+fw+1-f-vfw.$$

Expanding z^2 with $wv=1$ and $fv=wf=0$, every term cancels except 1, so $z^2=1$; likewise $zf=vf$. The factorization

$$\mathrm{diag}(z,z^{-1})=e_{12}(z)e_{21}(-z^{-1})e_{12}(z)\,e_{12}(-1)e_{21}(1)e_{12}(-1)$$

puts $h=\mathrm{diag}(z,z)$ in $\mathrm{EL}_2(R)$, and $he_{12}(fr)h^{-1}=e_{12}(zfrz)=e_{12}(v\,frz)\in N$. So $e_{12}(r)=e_{12}(vwr)\,e_{12}(fr)\in N$ for every $r\in R$, and conjugating by $e_{12}(1)e_{21}(-1)e_{12}(1)$ gives $e_{21}(R)\le N$. □

Proof of Theorem B. Write $e = 1 - st$. Then $e^2 = e$, $es = te = 0$, and there are finitely many $a_j, b_j \in R$ with $\sum_j a_j e b_j = 1$.

Rank four over a finitely generated ring. Suppose first that R is finitely generated as a unital ring, and put $G = \mathrm{EL}_4(R)$ and $L = \mathrm{EL}_3(R)$ on coordinates $1, 2, 3$. Both groups have property (T) by [EJZ, Theorem 1.1].

For $i = 1, 2, 3$, set

$$u_i = e_{4i}(t-1)e_{i4}(1)e_{4i}(s-1)e_{i4}(-t).$$

Its block on coordinates $(i, 4)$ is $\left(\begin{smallmatrix} s & e \\ 0 & t \end{smallmatrix}\right)$, so $u = u_3u_2u_1 \in \mathrm{EL}_4(R)$ is the matrix

$$u = \begin{pmatrix} s & 0 & 0 & e \\ 0 & s & 0 & et \\ 0 & 0 & s & et^2 \\ 0 & 0 & 0 & t^3 \end{pmatrix},$$

invertible as a product of elementary matrices. For $1 \le i \ne j \le 3$ and $a \in R$,

$$u\, e_{ij}(a) = e_{ij}(sat)\, u : \tag{5}$$

both sides differ from u by one entry, sa in position (i, j), since on the right the increment $sat \cdot (s, \ldots, et^{j-1})$ from row j of u contributes $sa\,ts = sa$ in position (i, j) and $sa\,te\,t^{j-1} = 0$ in position $(i, 4)$. So $uLu^{-1} \le L$.

The element

$$c = [e_{41}(e), e_{14}(t)] = \mathrm{diag}(1, 1, 1, 1 + et)$$

is computed in the block on coordinates $(1, 4)$ using $te = 0$, and a diagonal matrix of this shape commutes with every $\mathrm{diag}(A, 1)$, so $c \in C_G(L)$. Both uc and $e_{12}(e)u$ equal $u + et\,E_{14}$: the last column of uc is $(e, et, et^2, t^3)^{\mathsf{T}}(1 + et)$ and $e^2 = e$, $te = 0$, while $e_{12}(e)u = u + e\,(s, 0, 0, et)E_{1*}$ and $es = 0$. So

$$ucu^{-1} = e_{12}(e).$$

For $\ell = e_{23}(1)$ the Steinberg relation $[e_{12}(e), e_{23}(1)] = e_{13}(e)$ gives

$$d = [ucu^{-1}, \ell] = e_{13}(e) \in \mathfrak{D}_G(L).$$

Let N be the normal closure of d in G. For arbitrary $a, b \in R$, the Steinberg relations give

$$[e_{41}(a), e_{13}(e)] = e_{43}(ae), \qquad [e_{43}(ae), e_{32}(b)] = e_{42}(aeb).$$

It follows that $e_{42}(1) = \prod_j e_{42}(a_j e b_j) \in N$. Elementary signed permutation matrices conjugate this element to every off-diagonal position, up to a sign, so $e_{ij}(1) \in N$ whenever $i \ne j$. For distinct i, j, k and every $r \in R$,

$$[e_{ij}(1), e_{jk}(r)] = e_{ik}(r) \in N.$$

So $N = G$, and $\mathfrak{D}_G(L) = G$. By Theorem A with $K = G$, every homomorphism from G to an MF group is trivial.

Rank two. Now let R be arbitrary. By Lemma 3.1 choose $v_0, v_1, w_0, w_1 \in R$ with $w_iv_j = \delta_{ij}$, and let S be the unital subring they generate. It is finitely generated, $w_0v_0 = 1$, and $w_1(1 - v_0w_0)v_1 = 1$, so the first part applies to S: every homomorphism from $\mathrm{EL}_4(S)$ to an MF group is trivial. Put

$$S_1 = v_0v_0, \quad S_2 = v_0v_1, \quad S_3 = v_1v_0, \quad S_4 = v_1v_1,$$

$$T_1 = w_0w_0, \quad T_2 = w_1w_0, \quad T_3 = w_0w_1, \quad T_4 = w_1w_1,$$

so that $T_iS_j = \delta_{ij}$, and put $p = \sum_i S_iT_i$. For a 4×4 matrix A over S put

$$\jmath(A) = 1 - p + \sum_{i,j} S_iA_{ij}T_j \in R.$$

Since $(1-p)S_i = 0$ and $T_j(1-p) = 0$, we have $\jmath(AB) = \jmath(A)\jmath(B)$ and $\jmath(I) = 1$, so $\Psi(A) = \mathrm{diag}(\jmath(A), 1)$ defines a homomorphism $\mathrm{GL}_4(S) \to \mathrm{GL}_2(R)$, and for $i \neq j$ and $r \in S$,

$$\Psi(e_{ij}(r)) = \mathrm{diag}(1 + S_irT_j, 1) = [e_{12}(S_ir), e_{21}(T_j)]$$

by the computation in Lemma 3.2, since $T_jS_i = 0$. So $\Psi(\mathrm{EL}_4(S)) \leq \mathrm{EL}_2(R)$. Let ρ be a homomorphism from $\mathrm{EL}_2(R)$ to an MF group. Then $\rho \circ \Psi$ is trivial on $\mathrm{EL}_4(S)$, so ρ kills $\Psi(e_{12}(1)) = \mathrm{diag}(1 + S_1T_2, 1)$, which normally generates $\mathrm{EL}_2(R)$ by Lemma 3.2 with $v = S_1$, $w = T_1$, $a = S_2$, $b = T_2$. So ρ is trivial.

All ranks. For $n \geq 3$, a homomorphism from $\mathrm{EL}_n(R)$ to an MF group is trivial on the copy of $\mathrm{EL}_2(R)$ on coordinates $1, 2$, so it kills $e_{12}(R)$, and signed permutation matrices conjugate $e_{12}(r)$ to $e_{ij}(\pm r)$ for all $i \neq j$. □

Corollary 3.3. *If R is a countable simple unital ring that is not directly finite, then every homomorphism from $\mathrm{EL}_n(R)$ to an MF group is trivial for every $n \geq 2$. The same conclusion holds for $R = L_k(1, m)$, for every countable field k and every $m \geq 2$.*

Proof. In the first case choose $ts = 1 \neq st$; then $1 - st$ is a nonzero idempotent, and since R is simple it generates R as a two-sided ideal. In the second case, with generators $s_1, \ldots, s_m, t_1, \ldots, t_m$ subject to $t_is_j = \delta_{ij}$ and $\sum_i s_it_i = 1$, take $s = s_1$ and $t = t_1$; then $1 - s_1t_1 = \sum_{i \geq 2} s_it_i$ and

$$t_2(1 - s_1t_1)s_2 = 1.$$

Theorem B applies in both cases. □

Corollary 3.4. *If a countable unital ring R is not directly finite, then $C^*_{\max}(\mathrm{EL}_n(R))$ contains a proper isometry for every $n \geq 4$, and for every $n \geq 2$ if R satisfies the hypothesis of Theorem B. If $R \neq 0$ satisfies that hypothesis, then $C^*_{\mathrm{r}}(\mathrm{EL}_n(R))$ is separable, stably finite, and not MF for every $n \geq 2$, and the unit group $R^\times$ is not MF.*

Proof. Choose $s, t \in R$ with $ts = 1 \neq st$, and let S be the unital subring they generate. The compressor u from the proof of Theorem B lies in $\mathrm{EL}_4(S)$ and conjugates $L = \mathrm{EL}_3(S)$ into itself, and L has property (T) [EJZ, Theorem 1.1]. The compression is strict: every off-diagonal entry of a matrix in uLu^{-1} has the form sat, and $e(sat) = 0$ while $e \cdot 1 = e \neq 0$, so $e_{12}(1) \in L \setminus uLu^{-1}$. By Proposition 2.9 applied to $L \leq \mathrm{EL}_n(R)$ and u, $C^*_{\max}(\mathrm{EL}_n(R))$ contains a proper isometry. Under the hypothesis of Theorem B, take instead S, $L = \mathrm{EL}_3(S)$, u, and Ψ as in the rank-two part of its proof, with $s = v_0$ and $t = w_0$. Then Ψ is injective, since $T_i\jmath(A)S_j = A_{ij}$, so $\Psi(L) \leq \mathrm{EL}_2(R) \leq \mathrm{EL}_n(R)$ has property (T) and is compressed strictly by $\Psi(u)$, and Proposition 2.9 gives the proper isometry for every $n \geq 2$.

For the second assertion, $G = \mathrm{EL}_n(R)$ is countable and nontrivial, and by Theorem B it is not MF. The algebra $C^*_{\mathrm{r}}(G)$ is separable, and stably finite because its canonical trace is faithful. If an embedding $\iota \colon C^*_{\mathrm{r}}(G) \to \mathcal{Q}_{\mathbf{d}}$ existed and $p = \iota(1)$, then $g \mapsto$

$\iota(\lambda_g) + (1-p)$ would embed G in $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$, contradicting that G is not MF. Finally, $\jmath$ embeds $\mathrm{EL}_4(S)$ in $R^\times$; the group $\mathrm{EL}_4(S)$ is nontrivial with every homomorphism to an MF group trivial, so it is not MF, and restricting the maps V_n shows that MF passes to subgroups. So $R^\times$ is not MF. □

3.1. **The binary example.** Put $R = L_{\mathbb{F}_2}(1,2)$ and $H = R^\times$. By (3), the maps $x \mapsto (t_0x, t_1x)$ and $(y,z) \mapsto s_0y + s_1z$ are mutually inverse isomorphisms of right R-modules between R and $R \oplus R$, and $H \cong \mathrm{GL}_4(R) = \mathrm{EL}_4(R)$ [K–T, Proposition 4.2 and Corollary 4.4]. We identify H with $\mathrm{EL}_4(R)$.

Proof of Theorem C. The relations give $t_0s_0 = 1$ and $t_1(1-s_0t_0)s_1 = 1$, so R satisfies the hypothesis of Theorem B, and every homomorphism from H to an MF group is trivial. The ring R is finitely generated, so H has property (T) by [EJZ, Theorem 1.1], and hence is finitely generated [BHV, Theorem 1.3.1]. Since $e_{12}(1) \neq 1$, the group H is nontrivial.

Let N be a normal subgroup of H. The ring R is purely infinite simple [AA] and therefore an exchange ring [Ara], so by Preusser's sandwich theorem [Pre, Theorem 3] there is an ideal I of R with

$$\mathrm{EL}_4(R,I) \leq N \leq C_4(R,I),$$

where $\mathrm{EL}_4(R,I)$ is the normal closure in $\mathrm{EL}_4(R)$ of the elementary matrices $e_{ij}(a)$ with $a \in I$, and $C_4(R,I)$ is the preimage of the center of $\mathrm{GL}_4(R/I)$. Since R is simple, either $I = R$, and then $N \geq \mathrm{EL}_4(R) = H$, or $I = 0$, and then every $g \in N$ is central in $\mathrm{GL}_4(R)$. In the second case, commuting with each $e_{ij}(1)$ forces $g = \lambda I_4$ with $\lambda \in R^\times$, and commuting with each $e_{ij}(a)$ then gives $\lambda a = a\lambda$ for all $a \in R$. So λ lies in $Z(R) = \mathbb{F}_2$ [AC, Corollary 4.3], and $N = 1$. So H is simple, and in particular not MF.

Since $t_0s_0 = 1$ and $s_0t_0 \neq 1$, the ring R is not directly finite, so by Corollary 3.4 the algebra $C^*_{\mathrm{r}}(H)$ is separable, stably finite, and not MF, and $C^*_{\max}(H)$ contains a proper isometry. □

4. An amenable nonquasidiagonal trace

Brown observed that it was unknown whether every amenable trace on a C^*-algebra is quasidiagonal [Br, discussion preceding Proposition 3.5.1]. For a separable unital C^*-algebra A we use the sequential form of his definitions. A tracial state τ on A is *amenable* if there are u.c.p. maps $\phi_n \colon A \to M_{d_n}(\mathbb{C})$ such that

$$\|\phi_n(ab) - \phi_n(a)\phi_n(b)\|_2 \longrightarrow 0, \qquad \mathrm{tr}_{d_n}(\phi_n(a)) \longrightarrow \tau(a) \quad (a,b \in A).$$

It is *quasidiagonal* if the first limit holds in operator norm instead of normalized Hilbert–Schmidt norm, with the same trace convergence.

For a countable group G, let u_g denote the canonical unitary in $C^*_{\max}(G)$. The canonical trace τ_G is determined by

$$\tau_G(u_g) = \begin{cases} 1, & g = 1, \\ 0, & g \neq 1. \end{cases}$$

Theorem 4.1 (canonical-trace obstruction)**.** *Let G be a countable group. If G is not MF, then its canonical trace τ_G on $C^*_{\max}(G)$ is not quasidiagonal. In particular, if τ_G is amenable, then τ_G is an amenable trace that is not quasidiagonal.*

Proof. Suppose that τ_G is quasidiagonal, and let ϕ_n be u.c.p. maps as in the definition, with the first limit in operator norm. Then $\phi_n(u_g)^*\phi_n(u_g)$ and $\phi_n(u_g)\phi_n(u_g)^*$ converge to 1 in operator norm, so for large n the unitary part $V_n(g)$ of the polar decomposition of $\phi_n(u_g)$ satisfies
$$\|V_n(g) - \phi_n(u_g)\| \longrightarrow 0,$$
and $V_n(1) = 1$ because ϕ_n is unital. The maps $V_n \colon G \to \mathcal{U}(d_n)$ are asymptotically multiplicative in operator norm. If $g \neq 1$ and $\|V_n(g) - 1\| \to 0$, then $\mathrm{tr}_{d_n}(\phi_n(u_g)) \to 1$, contradicting $\mathrm{tr}_{d_n}(\phi_n(u_g)) \to \tau_G(u_g) = 0$. So $\limsup_n \|V_n(g) - 1\| > 0$ for every $g \neq 1$, and G is MF, contrary to the hypothesis. The last assertion follows. □

A group N is *locally residually finite* if each of its finitely generated subgroups is residually finite.

Proposition 4.2 (amenable extensions of locally residually finite groups)**.** *Let $1 \to N \to G \to A \to 1$ be an extension of countable groups with N locally residually finite and A amenable. Then the canonical trace of $C^*_{\max}(N)$ is quasidiagonal, and the canonical trace of $C^*_{\max}(G)$ is amenable.*

Proof. For the first assertion, let $E \subseteq N$ be finite, put $N_0 = \langle E \rangle$, and choose a finite quotient $\theta \colon N_0 \to Q$ with $\theta(x) \neq 1$ for $x \in E \setminus \{1\}$; put $H = \ker\theta$. In the quasi-regular representation of N on $\ell^2(N/H)$, the finite-dimensional subspace $\ell^2(N_0/H)$ is N_0-invariant, and compression to it is a u.c.p. map ϕ_E on $C^*_{\max}(N)$ that restricts to a representation of $C^*_{\max}(N_0)$. For $x \in E \setminus \{1\}$, the element x fixes no coset nH with $n \in N_0$, because $n^{-1}xn \in H$ would give $x \in H$ by normality of H in N_0; so $\mathrm{tr}(\phi_E(u_x)) = 0$. Along an exhaustion of N by finite sets E, the maps ϕ_E are eventually multiplicative on each pair of canonical unitaries, and their traces converge to τ_N on every canonical unitary; density of the group algebra and contractivity extend both limits to $C^*_{\max}(N)$.

For the second assertion, write $g \mapsto \bar{g}$ for the quotient map $G \to A$, and fix a finite set $E \subseteq G$, a finite Følner set $F \subseteq A$, and a section $\sigma \colon A \to G$. For $g \in E$ and $x \in F$ put
$$b(g, x) = \sigma(\bar{g}x)^{-1} g\sigma(x) \in N.$$
The subgroup N_0 generated by these finitely many elements is residually finite; choose a finite quotient $\theta \colon N_0 \to Q$ with $\theta(b(g, x)) \neq 1$ whenever $b(g, x) \neq 1$, put $H = \ker\theta$, and choose representatives $r_q \in N_0$ of the cosets of H in N_0. The cosets $\sigma(x)r_qH$, $x \in F$, $q \in Q$, are pairwise distinct, since equality forces $x = x'$ in A and then $q = q'$; let T be their set, let P be the projection of $\ell^2(G/H)$ onto $\ell^2(T)$, and put $\Phi(b) = P\lambda(b)P|_{\ell^2(T)}$ for $b \in C^*_{\max}(G)$, where λ is the quasi-regular representation. This map is u.c.p., and for $g \in E$ and $\bar{g}x \in F$,
$$g\,\sigma(x)r_qH = \sigma(\bar{g}x)\, r_{\theta(b(g,x))q}H \in T.$$

So for $g, h \in E$ the identity $\Phi(u_{gh}) - \Phi(u_g)\Phi(u_h) = P\lambda(g)(1-P)\lambda(h)P$ and the rank of $(1-P)\lambda(h)P$, at most $|\{x \in F : \bar{h}x \notin F\}|\,|Q|$, give

$$\|\Phi(u_{gh}) - \Phi(u_g)\Phi(u_h)\|_2^2 \le \frac{|\{x \in F : \bar{h}x \notin F\}|}{|F|}.$$

The normalized trace of $\Phi(u_g)$ is the fraction of points of T fixed by g, and for $g \in E \setminus \{1\}$ it is zero: if $\bar{g} \neq 1$, then g moves the F-coordinate of every point, and if $\bar{g} = 1$, then $g\sigma(x)r_qH = \sigma(x)r_{\theta(b(g,x))q}H$ with $b(g,x) = \sigma(x)^{-1}g\sigma(x) \neq 1$, so $\theta(b(g,x))q \neq q$. Taking Følner sets with $|\{x \in F : \bar{h}x \notin F\}|/|F| \to 0$ for the finitely many $h \in E$, along an exhaustion of G by finite sets E, the maps Φ are asymptotically multiplicative in normalized Hilbert–Schmidt norm and their traces converge to τ_G on the canonical unitaries; density and contractivity extend both limits to $C^*_{\max}(G)$. $\square$

We now construct W. For W to be non-MF, Γ may be any countable property-(T) group with a proper self-embedding; residual finiteness of Γ and finiteness of the index of the self-embedding are needed only for the trace (Proposition 4.4). Let Γ be a countable group with property (T), let $\alpha\colon \Gamma \to \Gamma$ be injective but not surjective, and choose $a \in \Gamma \setminus \alpha(\Gamma)$. Put

$$T_\alpha = \varinjlim(\Gamma \xrightarrow{\alpha} \Gamma \xrightarrow{\alpha} \cdots), \qquad V = T_\alpha \rtimes \langle t \rangle,$$

where $tgt^{-1} = \alpha(g)$ on the level-zero copy of Γ, so that V is the ascending HNN extension of Γ along α and $T_\alpha = \bigcup_{n\ge 0} t^{-n}\Gamma t^n$. Put $X = V/\Gamma$, the left-coset space. The Clifford lamp group $\mathrm{Cl}(X)$ is the group with generators ε and c_x, $x \in X$, and relations

$$\varepsilon^2 = c_x^2 = 1, \qquad [\varepsilon, c_x] = 1, \qquad [c_x, c_y] = \varepsilon \quad (x \neq y).$$

To see that $\varepsilon \neq 1$, fix a total order on X, let E be the $\mathbb{F}_2$-vector space of finitely supported functions $X \to \mathbb{F}_2$, and for $f, g \in E$ put $B(f,g) = \sum_{x>y} f(x)g(y)$. Since B is bilinear,

$$(a, f)(b, g) = \big(a + b + B(f,g),\ f + g\big)$$

is a group law on $\mathbb{F}_2 \times E$, in which $(1,0)$ is a central involution, each $(0, \delta_x)$ is an involution, and for $x \neq y$ exactly one of $B(\delta_x, \delta_y)$ and $B(\delta_y, \delta_x)$ equals 1, so the commutator of $(0,\delta_x)$ and $(0,\delta_y)$ is $(1,0)$. So $\varepsilon \mapsto (1,0)$, $c_x \mapsto (0,\delta_x)$ defines a homomorphism $\mathrm{Cl}(X) \to \mathbb{F}_2 \times E$. The relations let us write every element of $\mathrm{Cl}(X)$ as

$$\varepsilon^a c_{x_1} \cdots c_{x_r}, \qquad a \in \mathbb{F}_2, \quad x_1 < \cdots < x_r,$$

and this word maps to $(a, \delta_{x_1} + \cdots + \delta_{x_r})$, so the expression is unique and the homomorphism is an isomorphism. In particular, for a finite subset $Y \subseteq X$, the subgroup generated by ε and the c_y with $y \in Y$ has order $2^{|Y|+1}$, so $\mathrm{Cl}(X)$ is countable and locally finite. The relations are invariant under permutations of X, so every permutation of X induces an automorphism of $\mathrm{Cl}(X)$ that permutes the c_x accordingly and fixes ε. In this way V acts on $\mathrm{Cl}(X)$ through its action on X, and we put

$$W = \mathrm{Cl}(X) \rtimes V. \tag{6}$$

Proposition 4.3 (Clifford obstruction from a proper self-embedding)**.** *The group W is not MF. More precisely, every homomorphism from W to an MF group kills ε.*

Proof. Identify Γ with its level-zero copy in $V \leq W$, and let c be the lamp at the root coset $\Gamma \in X$; it centralizes Γ, which fixes that coset. Then tct^{-1} is the lamp at $t\Gamma$ and $a(tct^{-1})a^{-1}$ is the lamp at $at\Gamma$. These cosets are distinct, because $t\Gamma = at\Gamma$ would mean $a \in t\Gamma t^{-1} = \alpha(\Gamma)$. So

$$[tct^{-1}, a(tct^{-1})a^{-1}] = \varepsilon \neq 1. \tag{7}$$

Set

$$x = tct^{-1}, \qquad y = axa^{-1}, \qquad d = [x, a].$$

Both x and y are involutions, and hence

$$d = xaxa^{-1} = xy, \qquad d^2 = (xy)^2 = [x, y] = \varepsilon.$$

Since $t \in \mathrm{Comp}_W(\Gamma)$, $c \in C_W(\Gamma)$ and $a \in \Gamma$, the element $d = [tct^{-1}, a]$ is one of the generators of $\mathfrak{D}_W(\Gamma)$. Thus

$$\langle \varepsilon \rangle \leq \mathfrak{D}_W(\Gamma).$$

The subgroup $\langle \varepsilon \rangle$ is central, so normal, and it is finite, so it has property (T). So Theorem A applies to the countable group W with $L = \Gamma$ and $K = \langle \varepsilon \rangle$: every homomorphism from W to an MF group is trivial on ε. In particular, W is not MF. □

Proposition 4.4 (the locally residually finite case)**.** *Suppose in addition that Γ is residually finite and that $[\Gamma : \alpha(\Gamma)] < \infty$. Then*

$$W \cong K \rtimes \mathbb{Z}, \qquad K = \mathrm{Cl}(X) \rtimes T_\alpha,$$

where K is locally residually finite. Consequently W is sofic but not MF, the canonical trace on $C^*_{\max}(K)$ is quasidiagonal, and the canonical trace on $C^*_{\max}(W)$ is amenable and not quasidiagonal.

Proof. Reassociating the semidirect products yields

$$W \cong K \rtimes \mathbb{Z}, \qquad K = \mathrm{Cl}(X) \rtimes T_\alpha.$$

For $n \geq 0$, write $\Gamma_n = t^{-n}\Gamma t^n$ for the image of the nth copy of Γ in T_α. Since $[\Gamma : \alpha(\Gamma)] < \infty$, each Γ_n has finite index in Γ_{n+1}; every element of T_α lies in some Γ_m, and conjugation by t maps Γ_{m+1} onto Γ_m, so Γ is commensurated by V. Hence the stabilizer in Γ_n of every point of $X = V/\Gamma$ has finite index, and every Γ_n-orbit in X is finite.

A finite subset of K involves only finitely many lamps c_x, and its T_α-coordinates lie in one Γ_n. Let Y be the union of the Γ_n-orbits of these finitely many x. Then Y is finite and Γ_n-invariant, and the finite subset is contained in

$$C_Y \rtimes \Gamma_n,$$

where $C_Y = \langle \varepsilon, c_y : y \in Y \rangle$ is finite by the normal form above. This semidirect product is residually finite. Indeed, an element with nontrivial Γ_n-component survives in a finite quotient of Γ_n. For a nontrivial element of C_Y, let J be the kernel of the action $\Gamma_n \to \mathrm{Aut}(C_Y)$. The quotient Γ_n/J is finite, and the element has nontrivial image in the finite group $C_Y \rtimes (\Gamma_n/J)$. Therefore every finitely generated subgroup of K is residually finite.

Each finitely generated subgroup of K is residually finite and hence sofic. The group K is their directed union, and $W/K \cong \mathbb{Z}$ is amenable. Soficity passes to directed unions and to extensions with amenable quotient [ES, Theorem 1], so W is sofic. Proposition 4.3 shows that it is not MF. By Proposition 4.2, applied to the extension $1 \to K \to W \to \mathbb{Z} \to 1$, the canonical trace of $C^*_{\max}(K)$ is quasidiagonal and that of $C^*_{\max}(W)$ is amenable; by Theorem 4.1 the latter is not quasidiagonal. □

For a concrete instance, take

$$\bar{\Gamma} = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}) = \left\{ \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix} : A \in \mathrm{SL}_3(\mathbb{Z}),\ v \in \mathbb{Z}^3 \right\} \leq \mathrm{GL}_4(\mathbb{Z}).$$

It is residually finite, since reduction modulo a suitable integer separates any two distinct integral matrices, and it has property (T) [BHV, Example 1.7.4(i)]. Let

$$D = \mathrm{diag}(2, 2, 2, 1), \qquad \alpha(g) = DgD^{-1},$$

so that $\alpha(v, A) = (2v, A)$. This is injective, and its image consists of the affine matrices whose translation coordinates are all even, so $[\bar{\Gamma} : \alpha(\bar{\Gamma})] = 8$. Translation a by the first standard basis vector lies outside $\alpha(\bar{\Gamma})$. So the preceding construction applies to $\bar{\Gamma}$, α and a.

Proof of Theorem D. Apply Proposition 4.4 to $\bar{\Gamma}$, α, and a. □

5. A torsion-free finitely presented example

For an acylindrically hyperbolic group G, choose the generating set A provided by Hull [Hu, Theorem 3.12]. A subgroup is *suitable* if it acts non-elementarily on the Cayley graph $\mathrm{Cay}(G, A)$ and normalizes no nontrivial finite subgroup [Hu, Definition 1.4]. Hull states his small cancellation theorem [Hu, Theorem 7.1] with injectivity on a ball of $\mathrm{Cay}(G, A)$; since every finite subset of G lies in some ball, the theorem takes the following form.

Theorem 5.1 (Hull's small cancellation theorem)**.** *Let G be acylindrically hyperbolic, let $N \leq G$ be suitable with respect to A, and let $g_1, \dots, g_m \in G$ and a finite subset $\Omega \subseteq G$ be given. Then there is a surjective homomorphism $\varphi\colon G \to Q$ such that Q is acylindrically hyperbolic, $\varphi|_\Omega$ is injective, $\varphi(g_i) \in \varphi(N)$ for all i, and every element of finite order in Q is the image of an element of the same order in G.*

Hull's proof treats $m = 1$ by passing to $G/\langle\langle r \rangle\rangle_G$ for one element r and the general case by induction on m [Hu, proof of Theorem 7.1], so $\ker \varphi$ is the normal closure of m elements and Q is finitely presented when G is.

Lemma 5.2 (saturation)**.** *Let G be finitely presented, torsion-free, and acylindrically hyperbolic, let $N \trianglelefteq G$ be nontrivial, and let $\Omega \subseteq G$ be finite. Then there is a surjective homomorphism $\varphi\colon G \to Q$ such that Q is two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $\varphi|_\Omega$ is injective, and $\varphi(N) = Q$.*

Proof. The subgroup N is normal, and it is infinite because G is torsion-free and $N \neq 1$, so it acts non-elementarily on $\mathrm{Cay}(G, A)$ by Osin [Os, Lemma 7.1]; since G is torsion-free, N normalizes no nontrivial finite subgroup, so N is suitable. By Hull [Hu, Corollary 5.7 and Lemma 5.8], N contains two elements h_1, h_2 such that $N_0 = \langle h_1, h_2 \rangle$ is again suitable, possibly with respect to a larger generating set $A' \supseteq A$, and Theorem 5.1 holds for A' as well, since every finite subset of G lies in a ball of $\mathrm{Cay}(G, A')$. Apply Theorem 5.1 to N_0 with $g_1, \ldots, g_m$ a finite generating set of G and with the set Ω. Then $Q = \langle \varphi(g_1), \ldots, \varphi(g_m) \rangle \leq \varphi(N_0) \leq \varphi(N) \leq Q$, so $Q = \varphi(N_0) = \langle \varphi(h_1), \varphi(h_2) \rangle$ and $\varphi(N) = Q$. The quotient is torsion-free by Theorem 5.1 and finitely presented because $\ker \varphi$ is normally generated by m elements. □

Fournier-Facio constructs a finitely presented torsion-free group G_0 with property (T), a subgroup $\Gamma \leq G_0$ with property (T), an element $t \in G_0$ with $t\Gamma t^{-1} \leq \Gamma$, and a subgroup $J \leq G_0$ isomorphic to a finitely presented infinite simple group, such that $[\Gamma, J] = 1$ and $tJt^{-1} \leq \Gamma$ [FF, §2]. The group G_0 is obtained there as a common quotient of two finitely generated acylindrically hyperbolic groups by Hull's theorem [Hu, Corollary 7.4], which allows the quotient to be chosen acylindrically hyperbolic; we take G_0 to be such a quotient. Put $S = tJt^{-1}$. Since J is simple and nonabelian, J and S are perfect.

For $c \in J$ and $\ell \in S$, the commutator $[tct^{-1}, \ell]$ lies in $\mathfrak{D}_{G_0}(\Gamma)$ by (1), since t conjugates Γ into itself, c centralizes Γ, and $S \leq \Gamma$. As c ranges over J, the element tct^{-1} ranges over S, so $[S, S] \leq \mathfrak{D}_{G_0}(\Gamma)$, and S is perfect, so

$$S \leq \mathfrak{D}_{G_0}(\Gamma).$$

Proof of Theorem E. Let N be the normal closure of S in G_0; it is nontrivial because S is. By Lemma 5.2 applied to G_0, N, and $\Omega = \varnothing$, there is a surjective homomorphism $\varphi \colon G_0 \to Q$ with Q two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, and $\varphi(N) = Q$. The group Q is infinite because it is acylindrically hyperbolic, and it has property (T) as a quotient of G_0.

By (2), $\varphi(S) \leq \mathfrak{D}_Q(\varphi(\Gamma))$. The subgroup $\mathfrak{D}_Q(\varphi(\Gamma))$ is normal in Q, and the normal closure of $\varphi(S)$ is $\varphi(N) = Q$, so

$$\mathfrak{D}_Q(\varphi(\Gamma)) = Q.$$

Both Q and $\varphi(\Gamma)$ have property (T), as quotients of G_0 and Γ, respectively. By the last assertion of Theorem A, every homomorphism from Q to an MF group is trivial. If a quotient $\bar{Q}$ of Q is MF, then the quotient map $Q \to \bar{Q}$ is trivial, so $\bar{Q} = 1$. □

Corollary 5.3. *Let G be finitely presented, torsion-free, and acylindrically hyperbolic, and let $\Omega \subseteq G$ be finite. Then G has a two-generated, finitely presented, torsion-free, acylindrically hyperbolic quotient P with property* (T) *such that the quotient map is injective on Ω and every homomorphism from P to an MF group is trivial.*

Proof. Hull's common quotient theorem [Hu, Corollary 7.4], applied to G and to the group Q of Theorem E, gives a common quotient P that is acylindrically hyperbolic, with the map from G injective on a prescribed finite subset, since a torsion-free acylindrically hyperbolic group has no nontrivial finite normal subgroup.

For finitely generated inputs, the proof of that corollary consists of two applications of Theorem 5.1 to the free product $G * Q$, so P is torsion-free and finitely presented, as in the remark after Theorem 5.1. As a quotient of Q, the group P is two-generated and has property (T), and every homomorphism from P to an MF group pulls back to one from Q, so it is trivial. ☐

Corollary 5.4. *The algebra $C^*_{\mathrm{r}}(Q)$ is separable, unital, generated by two unitaries, simple, has a unique tracial state, has stable rank one, is stably finite, and is not MF.*

Proof. The group Q is countable, torsion-free, and acylindrically hyperbolic, so it contains a non-degenerate hyperbolically embedded subgroup [Os, Theorem 1.2] and has no nontrivial finite normal subgroup. Dahmani, Guirardel, and Osin give simplicity and uniqueness of the trace [DGO, Theorem 2.35], and Gerasimova and Osin give density of the invertible elements, which is stable rank one [GO, Theorem 1.1]. The algebra is separable and generated by the canonical unitaries of the two generators; its canonical trace is faithful, so it is stably finite; and an embedding into a norm matrix corona would embed Q into the corona's unitary group, against Theorem E. ☐

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